EE5203 L11 Worksheet

I2C and SPI - Basri Erdogan

Expected time: 30-45 minutes

Policy: Attempt every question before consulting the worked solutions.

Part A - What a clock buys

1. List the three things a UART cannot do that I2C can, and name the feature of I2C responsible for each.
2. A synchronous transfer can pause mid-byte for a second and then resume correctly; a UART frame cannot. Explain why, in terms of how each receiver decides where a bit is.
3. Complete the comparison.

	UART	I2C	SPI
Clock line			
Wires for 3 devices			
Addressing			
Reports a missing device			
Typical speed			

Part B - I2C addressing and framing

4. Draw the byte that goes on the wire when the master starts a **write** to device `0x27`. Label the address bits and the R/W bit, and give the hex value.
5. Complete the table.

Datasheet (7-bit)	HAL write (8-bit)	HAL read (8-bit)
`0x27`		
`0x68`		
`0x3C`		

6. A student passes `0x68` straight to HAL. Which device do they actually address, and what will the call return?
7. Explain ACK and NACK. Give one situation where the **master** sends a NACK deliberately, and say why.
8. Why can two devices pulling `SDA` at the same time not damage each other? Name the property responsible.
9. A team uses 10 kΩ pull-ups on a long bus and the signals look like slow ramps rather than square edges. Explain the cause and give the trade-off in choosing a smaller resistor.

Part C - I2C timing

10. Compute the time for one byte plus ACK at 100 kHz and at 400 kHz. Show the bit count you used.
11. The LCD backpack needs four I2C data bytes per character. Compute the time for one character and for a 16-character line, at both speeds.
12. Your project has a 1 ms tick. Choose a display refresh rate and defend it with numbers, stating what fraction of available time the display consumes at your chosen rate.
13. A teammate proposes refreshing the display every tick "because it is only a few bytes." Evaluate this with numbers and give the resulting CPU load.

Part D - SPI

14. Name the four SPI wires and their directions. Which one is not shared between devices, and why?
15. Explain what "full duplex" means for SPI, and why reading a register still requires transmitting something.
16. Complete the mode table.

Mode	CPOL	CPHA	Clock idles	Sampled on
0				
1				
2				
3				

17. A slave's timing diagram shows the clock idling **low**, with data sampled on the **falling** edge. Give CPOL, CPHA, and the mode number.
18. Explain what a loopback jumper proves, and — just as importantly — what it does **not** prove.
19. Your SPI reads return `0xFF` constantly. Give three distinct causes, and explain why the peripheral reports no error for any of them.
20. Why must `CS` stay low for a whole multi-byte transaction? Describe what typically happens if it is released between bytes.

Part E - Choosing and diagnosing

21. For each scenario, choose I2C or SPI and justify in one sentence.

Scenario	Bus	Why
Eight temperature sensors, one reading per second, four spare pins		
One SD card, streaming logged data		
A 16×2 character display		
A 320×240 colour display refreshed 30 times a second		

22. For each symptom, give the most likely cause and the first check.

Symptom	Likely cause	First check
Bus scan finds nothing at all		
Scan finds the device, writes fail		
Display lit but blank		
SPI loopback returns `0x00`		
SPI returns data shifted by one bit		

Exit self-assessment

- ☐ I can convert 7-bit 8-bit addresses without hesitating.
- ☐ I can compute I2C byte and line times at either speed.
- ☐ I can derive CPOL/CPHA from a timing diagram.
- ☐ I can explain what loopback proves and what it does not.
- ☐ I can justify a bus choice from device count and pin budget.